

# Prognosticating RULs while exploiting the future characteristics of operating profiles



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## ABSTRACT

We put forward a generic tool condition monitoring system capable of embracing the critical problem of prognosticating remaining useful life (RUL) under time-variant operating profiles witnessed in real-world production environments. First, we mathematically model the tool degradation progression via a new, adaptive, and hybrid stochastic degradation model, unifying (1) the real-time degradation signal characteristics from the sensor; (2) the rate of degradation characteristics from historical data; (3) the evolution of the future operating profile; (4) jerks owing to dynamic transitions. Next, for the account of reality, we formulated new mappings, i.e., degradation rate function, and jerk function. Successively, for the first time, we modeled the physics of the evolution of dynamic operating profiles for various scenarios. In the first scenario, the dynamic operating profile evolves in a deterministic way. Here, we approximate the evolution of future profiles as a piecewise constant function. Next, we embrace the accompanying scenario by aiding the future profile to be uncertain. This uncertainty in the evolution of the future profile is modeled via a discrete-time Markov chain. This is further extended to handle a more complex scenario where the prior understanding of the expected future profile is known. Though, the dynamic transitions of distinct profiles in the future are yet again subjected to uncertainty. We inventively capitalize on the total number of transitions in respective profiles and accordingly update the transition distribution. The updated distribution better characterizes the expected profile and eventually reduces the uncertainty. The resulting generalized system approximates the first passage time of the degradation process to a threshold and provides a precise life estimate in real-time. The system is extensively assessed via real-life vibration-based signals to substantiate the claim.

## 1. Introduction

Studies have exhibited that if computerized numerical control (CNC) machining systems are fortified with tool condition monitoring (TCM), it can cut down seventy-five percent of the downtime and upraise machine availability beyond fifty percent [1]. That being the case, significant efforts have been devoted to approximating cutting tool degradation progression and foreseeing the remaining useful life (RUL) [2, 3]. Although efforts in this field are significant, a realistic system to meet real-world industrial scenarios is still scarce [4, 5]. This is mainly owing to highly complex and time-variant operating profile of real machining systems, which make it hard to build a generalized system, robust enough to stand uncertainty in the evolution of dynamic operating profiles and reliable enough to show satisfactory performance under varied industrial scenarios (Note that operating profile refers to the specific combination of the levels of the significant operating

parameters viz. speed, feed, etc.). To date, the vast majority of TCM systems are constrained to a specific industrial scenario, monitored parameters, etc.

Moreover, they are strictly designed on the impression that along the entire lifespan of the cutting tool, the prevailing operating profiles are unvarying or does not affect the degradation [4, 6, 7]. It would be of practical value to equip the system with intelligence that allows responding to the time-variant operating profiles and adaptable under various real-world scenarios. Undeniably, it is challenging to cater to a realistic TCM system that is intelligent to lever the uncertainty of dynamic profiles. In other words, the invention of a generalized TCM system for dynamic operating profile and varied industrial scenarios is still open areas. Motivated by the discussion as mentioned earlier, the objective of this work is to address these challenges by offering a novel and a generic TCM system for a dynamic operating profile that enriches reliability of RUL predictions while meliorating applicability in diverse

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**Acronyms and abbreviations**

|      |                                |
|------|--------------------------------|
| CNC  | Computerized Numerical Control |
| TCM  | Tool Condition Monitoring      |
| RUL  | Remaining Useful Life          |
| RMS  | Root Mean Square               |
| DTMC | Discrete-Time Markov Chain     |
| F    | Feed                           |
| S    | Cutting Speed                  |
| D    | Depth of Cut                   |
| TTF  | Time-to-Failure                |
| LPE  | Lifetime Prediction Error      |
| PA   | Prediction Accuracy            |
| IS   | Industrial Scenario            |

**Notations**

|                  |                                                                         |
|------------------|-------------------------------------------------------------------------|
| $D_s(t)$         | Sensor-based degradation signal at time $t$                             |
| $D_s(0)$         | Deterministic current sensor-based observation                          |
| $\xi(t)$         | Operating profile at time $t$                                           |
| $Q$              | Set of operating profiles ( $Q = \{q_1, q_2, \dots, q_r\}$ )            |
| $q_r$            | Total number of operating profiles in the given set $Q$                 |
| $\omega$         | Degradation rate function                                               |
| $\omega(\xi(t))$ | Degradation rate function for an operating profile $\xi(t)$ at time $t$ |

|                       |                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $D_s(\xi(t_i))$       | Sensor-based degradation signal under a distinct operating profile $\xi(t_i)$ at time $t_i$                                                                 |
| $t_i$                 | Time at $i$ th instance                                                                                                                                     |
| $\chi_{A_i}$          | Indicator function over the interval $A_i$ at time instant $t_i$                                                                                            |
| $p_{ij}$              | Transition probability                                                                                                                                      |
| $\theta$              | Square matrix of dimension $q_r \times q_r$                                                                                                                 |
| $J_{ij}$              | Total number of times the operating profile $i$ switches over operating profile $j$ from time 1 to $N$ .                                                    |
| $\phi_n$              | Operating bins                                                                                                                                              |
| $\vartheta$           | Number of visits in the first operating bin ( $\phi_1$ )                                                                                                    |
| $G(\xi(t))$           | Jerk function                                                                                                                                               |
| $Z_j$                 | $j$ th transition time                                                                                                                                      |
| $U$                   | Jerk magnitude                                                                                                                                              |
| $\alpha$              | Average jerk magnitude                                                                                                                                      |
| $1^r(\xi(t))$         | Indicator function for jerk direction                                                                                                                       |
| $\wp(\omega(\xi(x)))$ | Indicator function of the rate of degradation while undertaking the evolution of the future operating profile as per the in-operation functioning structure |
| $Y(t)$                | Aggregate of operating profiles transition                                                                                                                  |
| $\varepsilon(t)$      | White noise stochastic process                                                                                                                              |
| $\sigma$              | Parameter of white noise stochastic process                                                                                                                 |
| $N$                   | Total number of observations                                                                                                                                |
| $T_{TTF}$             | Time for the signal to reach the failure threshold                                                                                                          |
| $RUL(t_i)$            | Remaining useful life at any point of time $t_i$                                                                                                            |
| $LT_a$                | Actual lifetime                                                                                                                                             |

real-world industrial scenarios.

Firstly, we devised a realistic mathematical framework to approximate the tool degradation progression via a new, adaptive, and hybrid stochastic degradation model. It is engineered to unite (1) the real-time degradation signal characteristics from the sensor; (2) the rate of degradation characteristics from historical data; (3) the evolution of the future operating profile; (4) jerks owing to dynamic transitions. This hybrid configuration as elements and their combinations plays a crucial role in capturing and approximating degradation progression under the dynamically evolving operating profiles. The other innovative feature is that in the model, the degree of divergence in tool's degradation is related to the severeness of the in-progress profile. Subsequently, we proposed a new sorting algorithm to order the profiles with regard to their impact on the corresponding degradation rate. Also, for the account of reality, new mappings, i.e., degradation rate function, and jerk function, are formulated.

Furthermore, we formulate pioneering adaptive functioning structures to incite generalization in diverse real-world industrial scenarios viz. degradation in (1) a deterministic, dynamic operating profile (appropriate for batch production environments, e.g., heavy engineering industries involved in the manufacturing of motors, switchgear, etc.); (2) a randomly-varying dynamic operating profile (appropriate for job production environments, e.g., aerospace and shipbuilding industries); (3) an expected but fluctuating future operating profile (prominently realistic for almost every micro to medium-scale production environments, e.g., manufacturing industries engaged in the production of spare-part, jig, and fixture for electronics, automobile, etc. industries). Withal, the model is correspondingly adaptive in perceiving the traditional degradation model, where the operating profiles remain the same along the lifespan, appropriate for mass production environments. The resultant generalized TCM system approximates the first passage time of the degradation process of cutting tools to a failure threshold and is used to deliver the RULs, in real-time. To finish, the validation of the proposed system is demonstrated via a real-life experimental case study employing vibration-based degradation signals from a high-speed CNC milling machining center. The verification results direct that the proposed system incisively predicts the RULs in all the real-world scenarios.

The novelty of this work is three-fold. The first is the innovative design of a generic TCM system that accounts for the future characteristics of the dynamic operating profiles while prognosticating RULs. It is grounded in the physics of degradation progression and is a function of operating profiles. As a result, the fundamental advantage of utilizing the proposed system to deal with time-variant operating profiles is its proficiency in communicating the future evolution of dynamic operating profiles instantaneously. Second is the consideration of all-encompassing cases of industrial scenarios. To the best of our knowledge, we are the first to think through a complex real-world scenario of expected but fluctuating future operating profiles. Third, it is not restricted to a specific machine tool, sensor, and so on; rather, the system is adaptive and can be rendered as a first universal perspective to TCM and, for that matter, any prognostics research. An additional contribution lies in the outcomes; extensive quantitative and qualitative performance investigations are carried out. Further, in contrast to the traditional approach, the implications of the offered system under different scenarios are experimentally examined. That magnifies the robustness and applicability of the offered system in diverse real-world production environments.

The remainder of this article is drafted as follows: the subsequent segment concisely surveys similar efforts. An in-depth formulation of the mathematical model is offered in [Section 3](#). In [Section 4](#), systematic performance investigations are demonstrated. Finally, [Section 5](#) concludes the work and contributions are highlighted.

## 2. Literature review

To foreground the contribution of this work and its position in the related research, we carry out a systematic literature review. In particular, we emphasize on two fundamental aspects that have traditionally been examined separately: (a) approximating the cutting tool degradation progression and predicting the RUL and (b) integrating the effects of operating profiles on tools deterioration.

A plethora of research focuses on approximating tool degradation progression and foreseeing the RUL [\[5, 7\]](#). Wherein, the mainstream of research is focused on the degradation signals derived solitary from a sensor or their combinations, which is correlated with tool wear/state.

For instance, a flank wear prediction system is formulated in [8], which is explicitly based on force signal analysis, proficient for precise wear prediction. However, the accurateness declined as the profiles were altered considerably. A framework using multi-sensors to enhance the reliability of monitoring under the static operating profile is proposed in [9]. A comprehensive review of sensors and executive strategies for TCM is carried out in [2, 3]. The mainline of these researches are focused on estimating the present wear or do cataloging as a healthy/faulty tool based on current signal observation. As a result, such approaches are inflexible towards RUL expectations, which does not aid the crucial function of TCM. Following that, in literature, close attention is paid to the RUL prediction of cutting tools. In this context, we highlight some significant contributions here [10–20]. However, even the utmost promising system is not certainly adaptable in real-world scenarios [7, 21], principally owing to inadequate generalization competencies viz. the usage is constrained to a particular operation/sensor/machine tool or solitary valid for specific industrial scenarios (predominantly, restricted to high volume of productions viz. mass production, where the operating profiles are time-invariant). Besides, none of these models account for the dynamic operating profiles and don't incorporate their effects on the degradation progression. In recent years, a few stochastic methods are designed. For instance, the random walk based method that estimates the RUL is demonstrated in [22]. Various stochastic degradation models gauging the failure distribution for cutting tools are compiled in [23].

Nevertheless, such frameworks address failure as a random incident and do not deliver statistics regarding the degradation progression peculiar to a tool operating in dynamic operating profiles. In essence, a broad cross-section of this literature assumes that the operating profile is unvarying or does not affect degradation [4, 6, 7]. Consequently, their applications are restricted in diverse industrial scenarios viz. batch or job production environments where the operating profiles are highly time-variant.

In distinction, the other radical of research emphases on exhibiting the effects of operating profiles on tool degradation or its manifestations. For instance, wear is predicted as a function of operating parameters viz. cutting speed, etc. in [24, 25]. An analytical approach to predict the tool wear pertaining to operating parameters viz. helix angle, etc. is presented in [26, 27]. These methods provide an excellent reference to model the effects of various operating profiles on tool wear. However, as these methods are not based on real-time measurements, their suitability is mainly limited to offline operating parameter optimization.

For the integrity of this extensive review, it is stated that scarce studies are accessible, associating degradation progression with the operating profiles. Although, they approach the problem in a way that whenever the degradation is approximated, the future operating profile is taken invariant and the same as the current profile. Most recent works where operating profiles effects were considered are from [28, 29]; they addressed the problem from an accelerated degradation testing perspective solely centered on the current offline observation at different operating levels. Though they do not model the physics

associated with the evolution of dynamic operating profiles. In contrast, the methodology offered in our work is conceptually unique as we explicitly address the challenges allied with time-variant operating profiles by integrating its physics, capturing the uncertainty in the evolution of dynamic operating profiles in real-time. Consequently, the approximated RUL taps past as well as the future characteristics of operating profiles.

### 2.1. Research gaps

Grounded on the decisive survey, we recognized critical research gaps, as follows:

Traditional TCM systems are designed on the impression that along the entire lifespan of the cutting tool, the prevailing operating profiles are temporally unvarying or do not significantly affect tool degradation. Although, in reality, the operating profiles frequently fluctuate with operating mode conversions, and mostly, exert significant effects on the degradation. Still, the impact of time-variant operating profiles on life estimation has not received enough consideration. Specifically, there is a need for a system that can prognosticate tool RULs while exploiting prior information, along with the future characteristics of dynamic operating profiles under diverse scenarios. In reality, a real-time TCM system designed for prognosticating RULs under time-variant industrial scenarios is still an open area that needs to be addressed.

Insufficient efforts are reported on the formulation of a generalized system that offers adaptive functioning structures, conforming to the challenges owing to different real-world scenarios viz. batch, job, and mass production environments. Consequently, there is a pressing demand for building a generic TCM system (not confined to a specific operation, industrial scenario, etc.) for strengthening the contemporary intelligent machining processes.

## 3. Methodology

A realistic mathematical modeling framework is devised via a new, adaptive, and hybrid stochastic degradation model, innovatively engineered to unite following critical roots of information:

### 3.1. The real-time degradation signal and rate of degradation characteristics

Fig. 1 displays real-life vibration-based degradation signals concerning root mean square (RMS) value from three similar cutting tools functioned under various profiles viz. low, medium, and severe operating profile. Note that the term “low operating profile” or “medium operating profile” refers to the operating profile's degradation inducing severity, which depends on the levels of a specific combination of the operating parameters viz. speed, feed, etc. For this example, the order of the operating profiles by severity is “low operating profile” < “medium operating profile” < “severe operating profile,” implying that low operating profile induces a smaller degradation than medium operating profile and severe operating profile induces more degradation

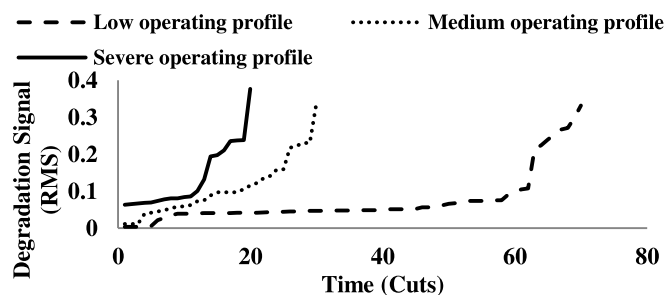


Fig. 1. Real-life degradation signals under various operating profiles.

than others (this sorting and ordering is briefed later in this sub-section). Fig. 1 suggests that the sensor-based degradation signals of even the same cutting tools operating under various operating profiles exhibit different functional forms. Given that, to capture the real-time degradation signal characteristic, the current observation of degradation signal from the in-situ sensor is incorporated in the proposed model. Moreover, Fig. 2 represents that under time-variant operating profile, the vibration level of the cutting tool increases as it transits from lower operating profile to severe operating profile or vice-versa. It is implied that the degree of divergence in the tool's degradation is related to the severeness of the in-progress operating profile. To induce this realistic characteristic in our model, we capitalize on the prior information from a population of the same cutting tools about the cutting tool's rate of degradation that it will follow under a specific operating profile. Consequently, we formulate a new degradation rate function and integrate it with the model.

$D_s(t)$  is any sensor-based degradation signal at time  $t$  and  $D_s(0)$  is deterministic current sensor-based observation. Moreover, let  $\xi(t)$  be any operating profile at time  $t$ , and the in-operation cutting tool can occupy one of the operating profile from a set of operating profiles  $Q$  ( $Q = \{q_1, q_2, \dots, q_r\}$ ,  $q_r$  is the total number of operating profiles in the given set of operating profiles  $Q$  and  $q_r < \infty$ ). Herein, determining the precise number of operating profiles, and a meaningful ordering of the operating profiles in  $Q$ , are essential aspects of our model. Although in general total numbers of profiles in which the manufacturing system will operate are known in advance. Accordingly, we undertake that the value  $q_r$  is identifiable. Following that, the degradation rate function is  $\omega(\omega : Q \rightarrow \mathbb{R}^+)$ . So,  $\omega(\xi(t))$  indicates tool's rate of degradation under a specific operating profile  $\xi(t)$  at time  $t$ , i.e., at any time  $\xi(t) = j \in Q$ , the cutting tool degrades at rate  $\omega(j)$  (we presume that  $\omega(j) > 0$  for each  $j \in Q$ ). (Note: rate of degradation is the change in degradation for a unit change in time along with life and is commonly projected from past historical data [30, 31].)

Next, we describe an essential aspect of the modeling framework, namely determining the set of operating profiles and arranging the elements of the set  $Q$  by ordering them according to their level of severity. In general, the degradation of cutting tools in any machining system is influenced by speeds, feeds, depth of cut, lubrication, humidity, etc. Though not all of the operating parameters are necessarily significant, and some of these possible combinations have a similar effect on the degradation rate. Since a vast number of operating profiles might raise possible computational issues, we focus on the scenario in which we choose the significant operating parameters (Note: significant operating parameters can be identified by number of ways viz. based on expert judgment [32], using a linear regression model [33], etc. We use expert judgment) and order the operating profiles. Let  $Z_1, Z_2, \dots, Z_{k'}$  represent  $k'$  significant operating parameters and  $\xi(m_1, \dots, m_{k'})$  denotes the profile when  $Z_1$  assumes level  $m_1$ ,  $Z_2$  assumes level  $m_2$ , and so forth. We now propose a sorting algorithm to order the profiles  $\xi(m_1, \dots, m_{k'})$  with regard to their impact on the corresponding degradation rate  $\omega(m_1, \dots, m_{k'})$ . We use observations from past historical data and perform a hypothesis test with the null hypothesis being  $H_0: \omega(m_1, \dots, m_{k'}) \geq \omega(m'_1, \dots, m'_{k'})$  against the alternate hypothesis

$H_1: \omega(m_1, \dots, m_{k'}) < \omega(m'_1, \dots, m'_{k'})$ . If there is sufficient evidence to reject the  $H_0$  at the  $\alpha$  level of significance, we conclude that profile  $\xi(m_1, \dots, m_{k'})$  is less severe than the profile  $\xi(m'_1, \dots, m'_{k'})$ . This comparison procedure can be applied pairwise to all profiles. So, we can order all of the operating profiles in  $Q$  from low to high by severity so that, for any two  $i, j \in Q$ ,  $i < j$  implies that profile  $i$  induces a smaller degradation rate than profile  $j$ .

### 3.2. The evolution of the future operating profile

Evident from Fig. 2, the transition among various operating profiles expressively contributes to the degradation progression. Consenting that, we need to reinforce our model to approximate the physics of evolution of time-variant operating profiles. Accordingly, for the first time, we formulate adaptive functioning structures to incite generalization in following diverse real-world scenarios:

#### 3.2.1. Industrial scenario I: a deterministic, dynamic operating profile

In this scenario, we emphasize on dynamic operating profiles that evolve in a deterministic manner. In this, the distinct operating profiles (say, low, medium, high, and severe) in the future are subject to change between each other. Though the times at which these operating profiles transits are known/deterministic, i.e., there is certainty about the transition times of distinct operating profiles, see, Fig. 3(a). Herein, the tool degradation changes when the tool is operated under diverse operating profiles, and such changes induce distinct degradation patterns that are usually significant, see Fig. 3(b). Such a scenario exceedingly arises in a batch production type of environments where the machining system runs at a particular profile to meet the requirement of a specific batch and transit to other profiles based on the prior batch scheduling decisions. Thus, the profile varies dynamically, but in a deterministic manner, as the distinct profiles operate for a specific time, as well as the time at which it transmits to other profile is also known/deterministic. Such a scenario is encountered in medium and heavy engineering industries (engaged in the manufacturing of electric motors, switchgear, machine tool, etc.), as these are allied with unique seasonal demand, or there is a requirement to manufacture diverse products.

The challenge lies in approximating the physics in arrears with future characteristics of the time-variant operating profiles that evolve in a deterministic manner. To go around this, the impact of in-progress operating profiles on the degradation rate is modeled, while undertaking the evolution of the future operating profile as a finite-valued deterministic and piecewise constant function. Accordingly, let  $\xi: [0, \infty) \rightarrow Q$  be a  $Q$ -valued piecewise constant function;  $\xi(t_i)$  is the operating profile at a discrete deterministic time  $t_i$  ( $i \in \{1, 2, \dots, T\}$ ). That is, in-operation the operating profiles in  $Q$  visits in a deterministic way. Following that, the proposed finite-valued deterministic and piecewise constant function  $\xi(t_i)$  is formulated as follows:

$$\xi(t_i) = \sum_{i=1}^T \alpha_i \chi_{A_i}(t_i) \text{ for all real numbers of } i. \quad (1)$$

where  $\alpha_i$  is a real number such that  $\alpha_i \in Q$  and  $\chi_{A_i}$  is the indicator

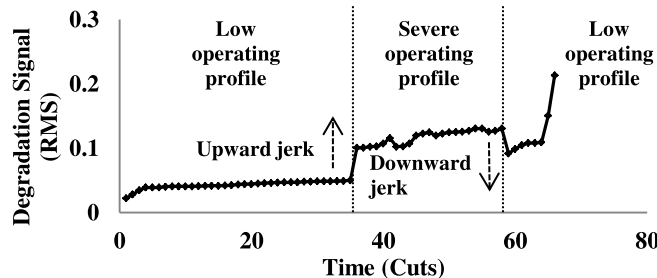


Fig. 2. Real-life degradation signal under time-variant operating profile.



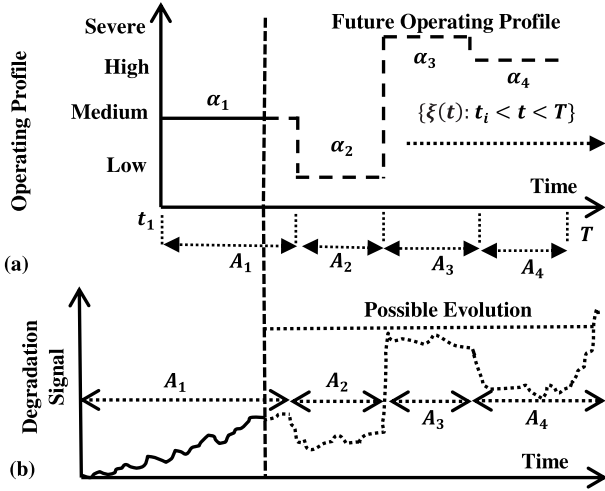


Fig. 3. Deterministic, dynamic operating profile.

function over the interval  $A_i$  at time instant  $t_i$ .

$$\chi_{A_i} = \begin{cases} 1 & \text{for } t_i \in A_i, \\ 0 & \text{for } t_i \notin A_i. \end{cases}$$

In this formulation, the intervals ( $A_i$ ) follows the following two properties:

- The intervals ( $A_i$ ) are pairwise disjoint set i.e.  $A_i \cap A_j = \emptyset$  for  $i \neq j$
- The union of the intervals covers the whole range of positive real numbers i.e.  $\cup_{i=1}^T A_i = \mathbb{R}^+$ .

Precisely, in accordance with Fig. 3,  $\xi(t_i)$  is expressed as:

$$\xi(t_i) = \begin{cases} \text{Medium } (\alpha_1), & t_i \in A_1 \\ \text{Low } (\alpha_2), & t_i \in A_2 \\ \text{Severe } (\alpha_3), & t_i \in A_3 \\ \text{High } (\alpha_4), & t_i \in A_4 \end{cases}$$

Finally, on the interval  $[t_i, T]$ , the previously formulated rate of degradation function while undertaking the evolution of the future operating profile as a finite valued deterministic and piecewise constant function can be formulated as,  $\int_{t_i}^T \omega(\xi(x))dx$ .

### 3.2.2. Industrial scenario II: a randomly-varying dynamic operating profile

The second scenario also takes up dynamic operating profiles but allows for the future operating profile to be uncertain. This means that the transition times in each distinct profile is exclusively random or time-variant, see Fig. 4. Such a scenario is exceptionally often in job production/one-off production environments, which take on the manufacturing of customized products, such as a one-time product for a specific customer or a small batch of products by clients' uncompromising demand. Herein, as each product is unique (varying in dimensions and material); it necessitates a distinct operating profile for machining. Thus, the operating profiles are dynamically varied in a manner to suit the uncertain requirements of a particular product. Aerospace and shipbuilding industries are recurrently lying open to such a scenario, as these are associated with high-quality goods, customized orders, and unique items.

Specifically, we need to generalize the functioning structure of scenario I by exploring a functioning structure where operating profiles vary randomly. Herein, the more prominent challenge lying ahead from before is in the approximation of physics in arrears with the inherent uncertainty of randomly-varying future operating profiles. To circumvent this, under periodically monitored situations, we undertake that

the randomly-varying dynamic operating profile progresses, rendering discrete-time Markov chain (DTMC). Accordingly, let  $\{\xi(t): t \geq 0\}$  be the DTMC and  $\xi(t)$  be the random operating profile. Herein, DTMC is for the set of operating profiles  $Q = \{q_1, q_2, \dots, q_r\}$ ,  $Q$  is discrete and have a finite number of operating profiles ( $q_r < \infty$ ). Herein,  $\{\xi(t): t \geq 0\}$  is a sequence of random operating profiles, say,  $(X_1, X_2, X_3, \dots, X_T)$  and is categorized by the memoryless property and given as:

$$Pr(X_{n+1} = x_{n+1} | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = Pr(X_{n+1} = x_{n+1} | X_n = x_n). \quad (2)$$

As we have already mentioned, the set of possible operating profiles  $Q$  of  $X_n$  (where  $n$  is the index for time) is finite and we call it state space of the chain. The chain transition from one operating profile (say,  $q_i, \forall i \in \{1, \dots, q_r\}$ ) to another operating profile (say,  $q_j, \forall j \in \{1, \dots, q_r\}$ ) with the transition probability (say,  $p_{ij}$ ) in one step, and given as:

$$p_{ij} = Pr(X_1 = q_j | X_0 = q_i). \quad (3)$$

subjected to  $0 \leq p_{ij} \leq 1$ , and  $\sum_{j=0}^{\infty} p_{ij} = 1$ .

Next, to represent the probability distribution of transitions from one operating profile to another we formulate a transition matrix (say,  $\theta = (p_{ij})_{i,j}$ ). Where  $\theta$  is a square matrix of dimension  $q_r \times q_r$ , where each element of position  $(i, j)$  represents the transition probability  $p_{ij}$ . In most practical cases, the transition probability matrix is unknown. In view of that, we present a method to approximate the transition probability matrix from the historical operating profile transition dataset. Let  $Y_T = (y_1, \dots, y_T)$  is a  $q_r$  state chain with the parameter to estimate being transition probability matrix and constraints being:  $0 \leq p_{ij} \leq 1$  and  $\sum_{j=0}^{q_r} p_{ij} = 1$ . Accordingly, first, we formulate the likelihood function, which is as follows:

$$p(Y_T; \theta) = p(y_1 | y_{n-1}; \theta) p(y_{n-1} | y_{n-2}; \theta) \dots p(y_1 | y_0; \theta) p(y_0; \theta). \quad (4)$$

Next, we formulate the log-likelihood function, as follows:

$$\log p(Y_T; \theta) = \sum_{k=1}^T \log p(y_k | y_{k-1}; \theta) \log p(y_0; \theta). \quad (5)$$

Substituting the value of likelihood function in (5) and further solving gives:

$$= \sum_{i=1}^{q_r} \sum_{j=1}^{q_r} J_{ij}(T) \log p_{ij} + \sum_{i=1}^{q_r} \delta(y_0 - i) \pi_0(i).$$

where  $J_{ij}$  is the total number of times the operating profile  $i$  switches over operating profile  $j$  from time 1 to  $T$ .

Then, subjected to constraints, taking the derivative and set it equal to zero as,  $\frac{d}{dp_j} \log p(Y_T; \theta) = 0$  yields the one-step transition probabilities  $p_{ij} = \frac{J_{ij}(T)}{\sum_{j=1}^{q_r} J_{ij}(T)} = \frac{J_{ij}(T)}{D_i(T)}$ . Where,  $D_i(T)$  is the number of visits in  $i$ . Hereafter, arranging one-step transition probabilities into a matrix gives  $\theta = \begin{bmatrix} p_{11} & \dots & p_{1q_r} \\ \vdots & \ddots & \vdots \\ p_{q_r 1} & \dots & p_{q_r q_r} \end{bmatrix}$ . The complete training data is used to approximate this matrix. Finally, the randomly-varying process  $\{\xi(t): t \geq 0\}$  is the future operating profile from time  $t_0$  up to some future time  $t = n$ . The previously formulated rate of degradation function

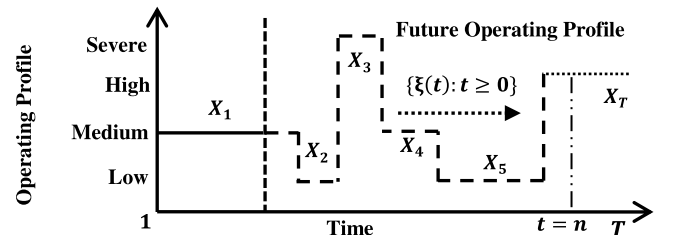


Fig. 4. Randomly-varying dynamic operating profiles.

while undertaking the evolution of the future operating profile as a discrete-time Markov chain can be formulated as,  $\int_0^t \omega(\xi(x))dx$ .

### 3.2.3. Industrial scenario III: an expected but fluctuating future operating profile

The third scenario addresses a greater extent of a complicated industrial scenario where the prior understanding of the expected future operating profile is standard or known in advance, withal; the dynamic transitions of operating profile in the future are subject to uncertainty. For instance, consider a manufacturing system operating under two distinct operating profiles (say, low and high), where it is expected/known that along the lifespan of the cutting tool, in the future, it will operate the total forty percent of the time at low operating profile, and remaining total sixty percent at the high operating profile. Though, there is existent uncertainty in the evolution of these standard operating profiles, as the future operating profile fluctuates randomly amongst distinct operating profiles along the lifespan of the cutting tool, see Fig. 5. Such a complicated scenario is prominently realistic for almost every micro to medium-scale production environments that are engaged in the production of multiple products, requiring distinct operating profiles. However, the demand forecasting may reveal the total expected duration under different profiles of the machining system. Spare-part, jig, and fixture manufacturing industries (engaged in the production of various standard spare-parts/fixtures for electronic, automobile, railway, etc.) are vastly exposed to such a scenario; as these are associated with standard parts with high variations in demands.

The more significant challenge existent here than before is in bringing together the restrictions (as per the expected future operating profiles) while approximating the physics in arrears with the evolution of uncertain fluctuating future operating profiles. To circumvent this complicated functioning structure of expected, but fluctuating operating profile. For the first time, we characterize discrete operating bins ( $\varphi_n$ ) for respective operating profiles and the percentage of the time the cutting tool will function in an individual operating bin is utilized as the additional statistics fed into the model while accounting the future evolution of expected but the fluctuating operating profile. Herein, we probabilistically induce a restriction in the evolution of the time-variant operating profile based on the expected future profile. In the interior, we designed the model to compute the aggregate of transitions in respective operating profiles; the instance the expected standard of the respective operational profile is achieved, the transition distribution is updated probabilistically. The updated distribution better characterizes the standard expected future profile and further reduces uncertainty. To mathematically illustrate this in accordance with Fig. 5, where, we consider a manufacturing system operating under two distinct operating profiles ( $Q = \{\text{Low (1), High (2)}\}$ ) with the known transition probabilities ( $\theta = \begin{matrix} \text{Low (1)} & 0.7 & 0.3 \\ \text{High (2)} & 0.5 & 0.5 \end{matrix}$ ). Whereas, it is expected that the system will operate 40% of the time in low operating profile and the remaining 60% in high operating profile. Accordingly, we set the operating bins as  $\varphi_1 = 40\% \xi(t)$  and  $\varphi_2 = 60\% \xi(t)$  and introduce a new set of states such as  $(1, \vartheta)$  and  $(2, \vartheta)$ , where,  $\vartheta$  is the number of visits in first operating bin ( $\varphi_1$ ). Subsequently, the transition probabilities of previously formulated DTMC with discrete operating bins ( $\{\xi(t): t \geq 0, \varphi_n\}$ ) changes in accordance with expected future operating profiles, and as follows:

$$Pr(\xi(t+1) = (1, \vartheta+1) | \xi(t) = (1, \vartheta)) = 0.7, \quad (6)$$

and

$$Pr(\xi(t+1) = (2, \vartheta) | \xi(t) = (1, \vartheta)) = 0.3. \quad (7)$$

Similarly,

$$Pr(\xi(t+1) = (1, \vartheta+1) | \xi(t) = (2, \vartheta)) = 0.5, \quad (8)$$

and

$$Pr(\xi(t+1) = (2, \vartheta) | \xi(t) = (2, \vartheta)) = 0.5. \quad (9)$$

As these transitions probabilities are formulated for the first operating bin ( $\varphi_1$ ),  $\vartheta \geq 40$ . Then, it turns into:

$$Pr(\xi(t+1) = 2 | \vartheta > 40) = 1. \quad (10)$$

In the same manner, any number of operating profiles can be modeled. Finally, the expected, but fluctuating process  $\{\xi(t): t \geq 0, \varphi_n\}$  is the future operating profile from time  $t_0$  up to some future time  $t$  with given expected discrete operating bins ( $\varphi_n$ ). The previously formulated rate of degradation function while undertaking the evolution of the future operating profile as expected, but the fluctuating process can be formulated as,  $\int_0^t \omega(\xi(x)) \varphi_n \% dx$ .

In essence, the impact in cooperation with the rate of degradation under all these diverse functioning structures on degradation progression is characterized and united in the proposed model.

### 3.3. Jerks owing to dynamic transitions

From Fig. 2, it is also prevalent that the vibration signal experiences an upward/downward jerk in the magnitude when the operating profile of the cutting tool transit from lower to higher or vice versa. Accordingly, for an account of reality, we think through that transition in operating profiles might bring upward/downward jerks in the magnitude. To bring this realistic characteristic in our model, we formulate a new mapping for jerks  $G: Q \rightarrow \mathbb{R}^+$  so that  $G(\xi(t))$  is a jerk function. Herein, the jerk magnitude is a random variable and can take any distribution. Following that, the impact of jerks owing to dynamic transitions is captured as follows:

$$G(\xi(Z_j)) = \mathcal{U} \times \mathbb{I}^r(\xi(t)). \quad (11)$$

where  $Z_j$  is  $j$ th transition time,  $\mathcal{U}$  is jerk magnitude which is modeled as a uniform random variable ( $\mathcal{U} \sim U(0, \alpha)$ ), with equal probability in the range of 0 and  $\alpha$ ,  $\alpha$  is the average jerk magnitude, estimated from training data),  $\mathbb{I}^r(\xi(t))$  is the indicator function for jerk direction (at particular operating profile transition times when the jerk is upward  $\mathbb{I}^r(\xi(t)) = "+"$ , and when the jerk is downward  $\mathbb{I}^r(\xi(t)) = "-"$ ).

In principle, this hybrid configuration of information captures and approximates degradation progression under dynamic operating profiles. Finally, the realistic mathematical model for sensor-based degradation signal progression approximation under dynamic operating profiles is given as follows:

$$D_s(t) = D_s(0) + \varphi(\omega(\xi(x))) + \sum_{j=1}^{Y(t)} G(\xi(Z_j)) + \sum_{t=1}^T \varepsilon(t). \quad (12)$$

where  $\varphi(\omega(\xi(x)))$  is the indicator function of the rate of degradation while undertaking the evolution of the future operating profile as per the in-operation functioning structure (for the scenario I:  $\varphi(\omega(\xi(x))) = \int_{t_i}^t \omega(\xi(x))dx$ , for scenario II:  $\int_0^t \omega(\xi(x))dx$ , and for scenario III:  $\int_0^t \omega(\xi(x))\varphi_n \% dx$ ),  $Y(t)$  is aggregate of operating profiles transition, and  $\varepsilon(t)$  is a white noise stochastic process.

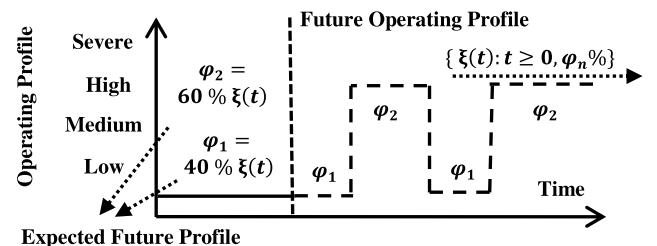


Fig. 5. Expected but fluctuating dynamic operating profiles.

The white noise stochastic process  $\varepsilon(t) \sim N(0, \sigma)$ , such that it is statistically independently and identically distributed at different time points. Thus, (12) can be re-written as:

$$D_s(t) = D_s(0) + \int_0^t \omega(\xi(x)) dx + \sum_{j=1}^{Y(t)} G(\xi(Z_j)) + \varepsilon(t). \quad (13)$$

The method for estimating the unknown parameter ( $\sigma$ ) of the white noise stochastic process from real-life data utilizing maximum likelihood estimation technique is presented in Appendix A. Experimental illustrations demonstrating the quality of this estimate is offered in the next section.

Our ultimate objective is prognosticating cutting tool RULs based on the current observations of any sensor signal process  $D_s(t)$  overtime, while exploiting prior information, along with the future characteristics of operating profiles, in real-time. Following that, the system approximates the first passage time of the degradation process to a failure threshold, i.e., the time-to-failure (TTF) and provides a precise life estimate in real-time. Thus, at any point of time  $t_i$ , the RUL can be algebraically calculated as follows:

$$RUL(t_i) = T_{TTF} - t_i. \quad (14)$$

where  $t_i$  is the point (time) of prediction,  $T_{TTF}$  is the time for the signal to reach the failure threshold (TTF).

Nevertheless, the proposed mathematical framework is correspondingly adaptive in distinguishing the traditional degradation model, appropriate for mass production environments. Subsequently, this generic TCM system will lead to a more precise estimate of RUL of the fielded cutting tool operating under dynamic operating profile.

To summarize, Fig. 6 illustrates the flow chart of the proposed methodology. Firstly, in the degradation data collection phase, condition monitoring data from sensors are collected. This is then transformed into sensor-based degradation signals. Lastly, a historical degradation dataset (i.e., multiple tools operated at different operating profiles) is constructed.

Secondly, in the degradation model development phase, model parameters viz.  $\omega(\xi(t)), \alpha, \sigma$ , etc. are estimated from the historical degradation dataset. On top, significant operating parameters are identified, and operating profiles are sorted based on severity. This information is then used to train the new, adaptive, and hybrid stochastic degradation model. Model's mathematical expression under different industrial scenarios is summarized as follows:

For Industrial Scenario I: A deterministic, dynamic operating profile.

$$D_s(t) = D_s(0) + \int_{t_i}^t \omega(\xi(x)) dx + \sum_{j=1}^{Y(t)} G(\xi(Z_j)) + \sum_{t=1}^T \varepsilon(t) \quad (15)$$

$\int_{t_i}^t \omega(\xi(x)) dx$ : rate of degradation function while undertaking the evolution of the future operating profile as a finite valued deterministic and piecewise constant function on the time interval  $[t_i, T]$ .

For Industrial Scenario II: A randomly-varying dynamic operating profile.

$$D_s(t) = D_s(0) + \int_0^t \omega(\xi(x)) dx + \sum_{j=1}^{Y(t)} G(\xi(Z_j)) + \sum_{t=1}^T \varepsilon(t) \quad (16)$$

$\int_0^t \omega(\xi(x)) dx$ : rate of degradation function while undertaking the evolution of the future operating profile as a randomly-varying process from time  $t_0$  up to some future time  $t$ . Herein, this  $\{\xi(x): x \geq 0\}$  discrete-time Markov chain approximates the evolution of future operating profiles.

For Industrial Scenario III: An expected but fluctuating future operating profile.

$$D_s(t) = D_s(0) + \int_0^t \omega(\xi(x)) \varphi_n \% dx + \sum_{j=1}^{Y(t)} G(\xi(Z_j)) + \sum_{t=1}^T \varepsilon(t) \quad (17)$$

$\int_0^t \omega(\xi(x)) \varphi_n \% dx$ : rate of degradation function while undertaking the evolution of the future operating profile as expected, but the fluctuating process from the time  $t_0$  up to some future time  $t$  with given expected discrete operating bins ( $\varphi_n$ ). Herein, this  $\{\xi(x): x \geq 0, \varphi_n\}$  expected, but the fluctuating process approximates the evolution of future operating profiles.

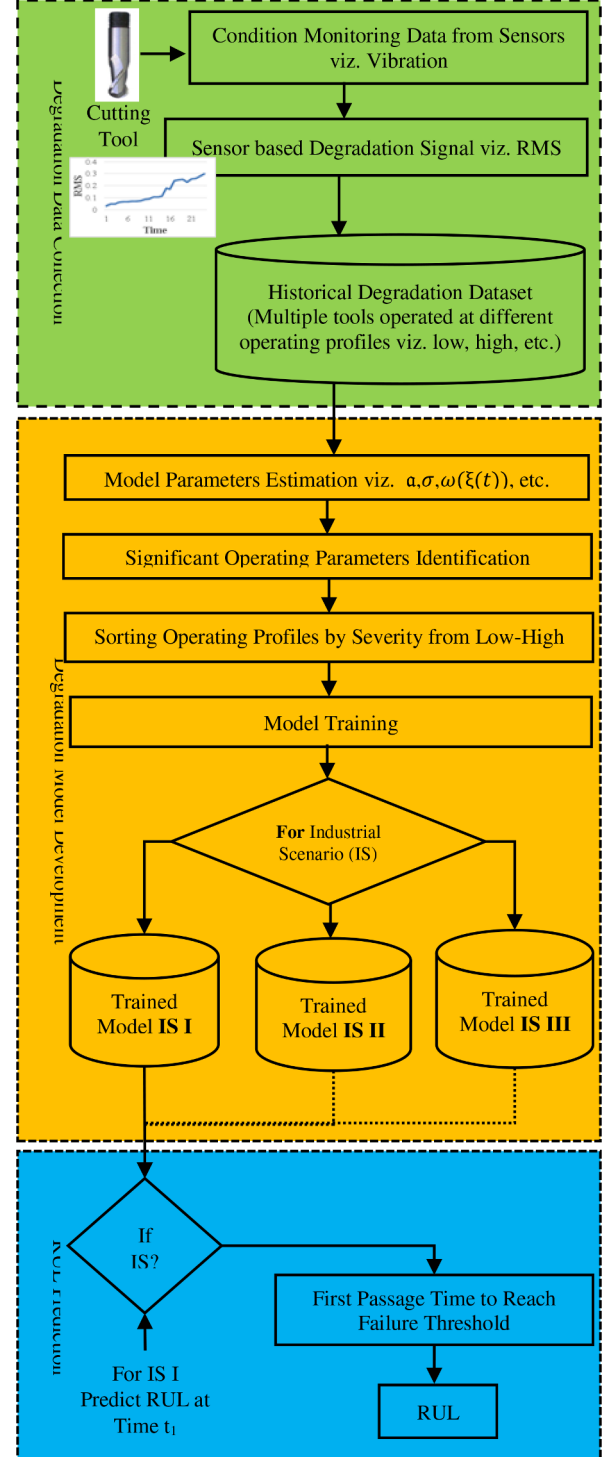


Fig. 6. Flow chart of the proposed methodology.

Lastly, in RUL prediction phase, the trained model is used for prediction at any given point of time. For instance, predicting RUL under the industrial scenario I at time  $t_1$ ; the model trained for the industrial scenario I get activated. The detailed mathematical expression for the scenario I is given in (15) and the approximation process is as follows:

The expression  $D_s(t)$  approximates the value of sensor-based degradation signal at the future time  $t_2$ . This is the future value of degradation signal when the tool reaches the time  $t_2$  from time  $t_1$ . This is the output of the model.

The expression  $D_s(0)$  is the value of sensor-based degradation signal at the current time  $t_1$  from where we want to predict the RUL of the tool. This is the input of the model.

The expression  $\int_0^{t_2} \omega(\xi(x)) dx$  approximates the evolution of the future operating profile. From this, the model learns that, when the tool reaches at a future time  $t_2$  from current time  $t_1$ , what will be the tool's operating profile and its rate of degradation. Only this expression changes for other scenarios rest of the model remains the same, for scenario II it is  $\int_0^{t_2} \omega(\xi(x)) dx$  (see, (16)); for scenario III it is  $\int_0^{t_2} \omega(\xi(x)) \varphi_n \% dx$  (see, (17)), respectively.

The expression  $\sum_{j=1}^{Y(t)} G(\xi(Z_j))$  maps jerks (triggers only when operating profiles transits); say if at time  $t_1$  the tool was operating at a low operating profile and in future at the time  $t_2$  tool transit to a high operating profile. The jerk function will trigger. Herein, if tool switches from low to high operating profile an upward jerk will take place otherwise downward and its magnitude is a uniform random variable  $U(0, \alpha)$ .

The expression  $\sum_{t=1}^T \varepsilon(t)$  approximates the noise in the degradation signal.

Following this, we get the value of  $D_s(t)$  at time  $t_2$ , now this becomes the input ( $D_s(0)$ ) and the value of  $D_s(t)$  is estimated for the future time  $t_3$  in the same manner as before. This runs perpetually until the model approximates the first passage time of the degradation process to a failure threshold. Once the degradation signal value ( $D_s(t)$ ) reaches failure threshold RUL is calculated using (14). Say, the failure threshold is reached at the time  $t_{50}$ ; using (20) RUL will be "failure time – current time"; i.e.  $t_{50} - t_1$  ( $50 - 1 = 49$ ).

## 4. Results and discussions

### 4.1. Experimental case study

An experimental case study is exhibited to demonstrate the functionality and practicality of the system.

#### 4.1.1. Experimental platform

An experimental platform is developed using the CNC milling machine as the test bench. The tool employed is a high-speed steel 6 mm

diameter flat end mill cutter. The workpiece material selected is a flat bar (165 mm × 100 mm × 20 mm) of mild steel. Machining operation engaged is face milling to form a flat plane surface of the workpiece. In each experiment, the cutter was used to cut the workpiece surface in succession to machine an entire surface of 1320 mm. All tests were performed under dry cutting conditions to accelerate tool degradation. Kistler accelerometer is mounted on the workpiece to measure the vibrations of the cutting process in the feed direction for the entire life of the cutters. The vibration signals were acquired with a frequency range from 1 to 10 kHz and amplified via Kistler coupler. Then, the signal is acquired by National Instruments data acquisition (DAQ) card with 2500 Hz sampling frequency. Moreover, the RMS value is recorded for each cut to record the degradation process. All cutters were run-to-failure, till breakage or till the RMS value reaches a pre-defined level ( $F_r = 0.30$ , as per engineering criteria based on equipment vibration, for example, the international organization for standardization (ISO) 2372). In contrast to the existing literature, the defects were not seeded instead of various defects (viz. breakage, etc.) were observed under run-to-failure tests, allowing a more realistic representation of TCM.

#### 4.1.2. Design of experiments

The L4 orthogonal array of the Taguchi method with six replications is chosen to adequately capture the rate of degradation characteristics from multiple combinations of distinct operating parameters. Specifically, three significant parameters with two levels are used, namely feed (F) (F: 250, 350 mm/min), cutting speed (S) (S: 1050, 1300 revolutions per minute (RPM)) and depth of cut (D) (D: 0.20, 0.35 mm). Four experimental runs derived from the L4 design are shown in Table 1 (Exp. ID 1–4). The rate of degradation characteristics of real-time vibration data throughout its life is taken as a measured response. Next, the distinct profiles are ordered according to their level of severity using the proposed sorting algorithm; see the severity column of Table 1, Exp. ID 1–4.

#### 4.1.3. Constructing signals with time-variant profiles

**Industrial Scenario I:** For constructing the deterministic, dynamic operating profile, we considered a manufacturing system operating under two distinct profiles (low and high), that evolves in a deterministic manner. So, we changed the operating profiles among these levels at known time intervals. The cutters were initially operated at the low profile, then just after sixteen cuts, the cutter transits to high profile till twenty-six cuts, and then transmit back to lower profile till failure.

a) **Industrial Scenario II:** For the randomly-varying, dynamic operating profile, we considered a manufacturing system is functioning in two profiles (medium and severe), that progresses stochastically. Given that, we randomly changed the profiles among these levels at uncertain time intervals. Explicitly, the cutters were primarily functioned at the severe profile, which randomly transits to medium profile, and then again randomly transits to severe profile till

**Table 1**  
Experimental design matrix.

| Constant operating profile<br>Exp. ID | Operating parameters                                                    | No. of Tools | Severity (State)                 |
|---------------------------------------|-------------------------------------------------------------------------|--------------|----------------------------------|
| E1                                    | (F:250, S:1050, D:0.20)                                                 | 6            | Low (1)                          |
| E2                                    | (F:350, S:1300, D:0.20)                                                 | 6            | Medium (2)                       |
| E3                                    | (F:250, S:1050, D:0.35)                                                 | 6            | High (3)                         |
| E4                                    | (F:350, S:1300, D:0.35)                                                 | 6            | Severe (4)                       |
| Dynamic operating profile             |                                                                         |              |                                  |
| E5                                    | (F:250, S:1050, D:0.20)→(F:350, S:1050, D:0.35)→(F:250, S:1050, D:0.20) | 2            | Industrial scenario              |
| E6                                    | (F:250, S:1300, D:0.35)→(F:350, S:1300, D:0.20)→(F:250, S:1300, D:0.35) | 2            | Deterministic                    |
| E7                                    | (F:250, S:1050, D:0.20)→(F:250, S:1300, D:0.35)→(F:250, S:1050, D:0.20) | 2            | Randomly-varying                 |
| E8                                    | (F:250, S:1050, D:0.20)→(F:350, S:1050, D:0.35)→(F:250, S:1050, D:0.20) | 1            | Expected but fluctuating         |
| E9                                    | (F:250, S:1300, D:0.35)→(F:350, S:1300, D:0.20)→(F:250, S:1300, D:0.35) | 1            | Deterministic                    |
| E10                                   | (F:250, S:1050, D:0.20)→(F:250, S:1300, D:0.35)→(F:250, S:1050, D:0.20) | 1            | Randomly-varying dynamic profile |
|                                       |                                                                         |              | Expected but fluctuating         |



failure.

- b) Industrial Scenario III: For the expected but fluctuating dynamic operating profiles, we considered a manufacturing system operating under two profiles (low and severe), where it is expected that the system will operate approximately 45% of the time in low profile and remaining 55% in severe profile. So, we retained the profile expectancy and randomly changed the profiles among these levels at uncertain time intervals. In detail, we set the operating bins as  $\varphi_1 = 45\% \xi(t)$  and  $\varphi_2 = 55\% \xi(t)$  and then, first, the cutters were operated at the low profile, which randomly transits to severe profile, while roughly accounting the profile expectancy, the profile again randomly interchanged to low profile till failure. Note the expected standard of the profile was maintained.

Following all these scenarios, we further conducted two groups of accelerated life tests (see Table 1, Exp. ID 5–10). In the first set Exp. ID 5–7 each cutter is operated with profiles rendering different real-world scenarios. By this set and along with the Exp. ID 1–4, we estimate the model parameters. The second set Exp. ID 8–10 is used for online validation under diverse scenarios. This design maximizes the likelihood that sufficient real-time data is collected to verify the performance under dynamic operating profiles.

#### 4.1.4. Experimental validation under diverse scenarios

For validating and assessing the performance of the system under the scenario I, data from Exp. ID 1, 3, and 5 are used for training and data from Exp. ID 8 is used for online validation. While, for scenario II, data from Exp. ID 2, 4, and 6 are used for training and data from Exp. ID 9 is used for online validation.

Similarly, for scenario III, data from Exp. ID 1, 4, and 7 are used for training and data from Exp. ID 10 is used for online validation. Table 2 offers the entire model parameters estimated from training data. Next, to meticulously validate the performance and to investigate the influence of dynamic operating profiles on the RUL prediction results, a comprehensive qualitative and quantitative performance investigation is carried out, as follows:

- a) *Qualitative performance investigation*: We begin with examining the dynamic operating profile evolution path (under uncertainty) approximated by the system for validation cutting tools (see Figs. 7 and 8). It can be manifested that the system is competent in approximating the uncertainty imposed by time-variant industrial scenarios. More prominently, in scenario III, the system approximated evolution path is virtually matching the actual evolution path (see, Fig. 8), as the updated distribution better characterizes the expected future profile and eventually reduces the uncertainty. Next, the validation tools were engaged to predict the RULs under diverse scenarios at different intervals of the lifetime percentiles. The predicted RULs were estimated from  $q_i^{th} - q_j^{th}$  percentiles ( $q_i - q_j \in \{0 - 30\%, 31 - 60\%, 61 - 90\%\}$ ). (Note: a lifetime percentile indicates the lifetime value below which a given percentage of lifetime observations in a group of lifetime

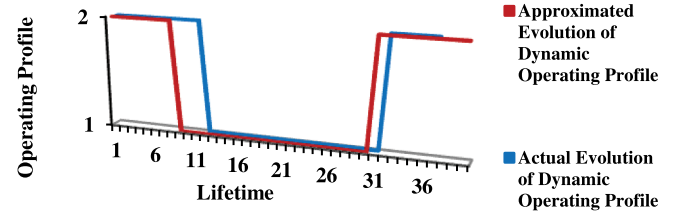


Fig. 7. Approximated and actual evolution path for scenario II.

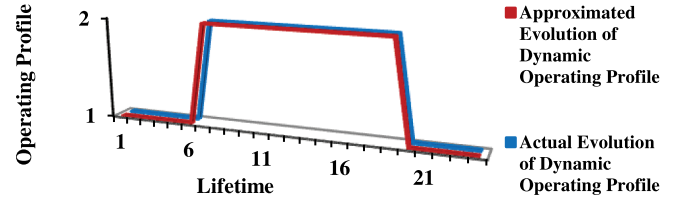


Fig. 8. Approximated and actual evolution path for scenario III.

observations falls. For instance, say a tool in its lifetime made 40 cuts. The 30th percentile of the tools lifetime will be  $\cong 12$  cut. Thus, the range 0th–30th percentile will include all the time-steps between 0th and 30th percentiles, i.e., 1 to 12 cuts from where RULs will be predicted.) Herein, the measure of performance is the lifetime prediction error (LPE).

$$LPE = \left| 1 - \frac{\text{Current Observation Epoch} + RUL_{p_i}}{LT_a} \right| \quad (18)$$

where  $RUL_{p_i}$  is the predicted RUL, ( $\text{Current Observation Epoch} + RUL_{p_i}$ ) is the predicted lifetime, and  $LT_a$  is the actual lifetime.

Figs. 9–11 show the prediction performance allied to a different interval of lifetime percentile under diverse scenarios. These figures revealed a significant finding that the prediction error under diverse scenarios exhibits progressive reduction for the most part as we move towards the end of the lifetime of the tool (i.e., as  $q_i^{th} - q_j^{th}$  increases). These promising results in time-variant scenarios guarantee the expansion of an effective predictive maintenance plan.

#### a) Comparative analysis of quantitative performance

We carried out a comprehensive quantitative performance investigation in contrast to the extensively adopted traditional approach or way of doing prognostics [34]; see Table 3 for detailed results. In general, the traditional approach/way means; an RUL prediction model designed on the impression that along the entire lifespan of the system, the prevailing operating profiles are unvarying or do not affect the degradation. So, the proposed model is adapted to perceive the traditional approach/way on the same set of experimental data.

**Table 2**  
Estimated model parameters from training data.

| Industrial scenario I         |                                                                                                                     | Industrial scenario II          |                                                                                                    | Industrial scenario III                                            |                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Parameter                     | Value                                                                                                               | Parameter                       | Value                                                                                              | Parameter                                                          | Value                                                                                           |
| $Q$                           | {Low(1), High(2)}                                                                                                   | $Q$                             | {Medium(1), Severe(2)}                                                                             | $Q$                                                                | {Low(1), Severe(2)}                                                                             |
| $\xi(t_i)$                    | $\begin{cases} \text{Low}, & 0 \leq t < 16 \\ \text{High}, & 16 \leq t < 26 \\ \text{Low}, & 26 \leq t \end{cases}$ | $\{\xi(t): t \geq 0\}; \theta$  | $\begin{matrix} \text{Severe(1)} & [0.95 \ 0.05] \\ \text{Medium(2)} & [0.06 \ 0.94] \end{matrix}$ | $\{\xi(t): t \geq 0, \varphi_1 = 45\%, \varphi_2 = 55\%\}; \theta$ | $\begin{matrix} \text{Low(1)} & [0.90 \ 0.11] \\ \text{Severe(2)} & [0.08 \ 0.92] \end{matrix}$ |
| $\omega(\xi(\text{Low}(1)))$  | $7.28 \times 10^{-3}$                                                                                               | $\omega(\xi(\text{Medium}(1)))$ | $8.97 \times 10^{-3}$                                                                              | $\omega(\xi(\text{Low}(1)))$                                       | $7.28 \times 10^{-3}$                                                                           |
| $\omega(\xi(\text{High}(2)))$ | $1.23 \times 10^{-2}$                                                                                               | $\omega(\xi(\text{Severe}(2)))$ | $1.39 \times 10^{-2}$                                                                              | $\omega(\xi(\text{Severe}(2)))$                                    | $1.39 \times 10^{-2}$                                                                           |
| $\sigma$                      | $3.15 \times 10^{-2}$                                                                                               | $\sigma$                        | $0.019 \ 1.90 \times 10^{-2}$                                                                      | $\sigma$                                                           | $3.59 \times 10^{-2}$                                                                           |
| $\alpha$                      | $1.54 \times 10^{-2}$                                                                                               | $\alpha$                        | $3.00 \times 10^{-2}$                                                                              | $\alpha$                                                           | $3.00 \times 10^{-2}$                                                                           |

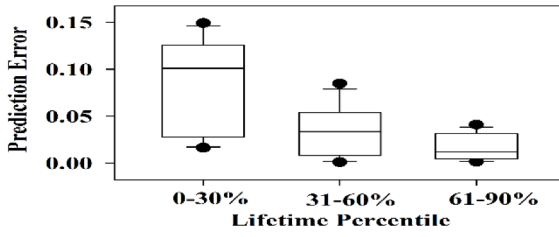


Fig. 9. Performance at different intervals of percentile for the scenario I.

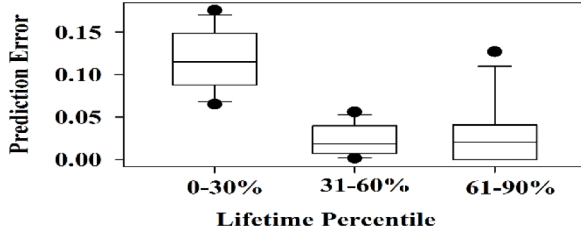


Fig. 10. Performance at different intervals of percentile for scenario II.

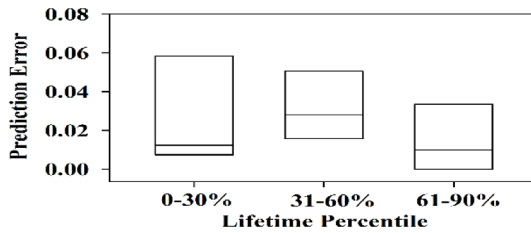


Fig. 11. Performance at different intervals of percentile for scenario III.

To better reflect the advantages of the proposed approach over the traditional approach, see Fig. 12(a). Herein, consider a tool operating in medium operating profile say  $\alpha_1$ ; and in future, it is going to operate dynamically under diverse operating profiles say,  $\alpha_2$ ,  $\alpha_3$ , etc. If we predict RUL at any point of time say,  $t_5$ ; the traditional approach while approximating the evolution of degradation signal from point  $t_5$  consider that tool will operate in the same profile ( $\alpha_1$ ) for the entire life-span of the tool. Thus, it does not incorporate the effect induced by future operating profiles. Though such changes induce distinct degradation patterns that are usually significant, see the actual degradation signal in Fig. 12(b). Whereas, the critical characteristic of the proposed approach is that while approximating the evolution of degradation signal from point  $t_5$ , it also approximates the evolution of future operating profiles via. proposed functioning structures. Thus, incorporate the effect of dynamic future operating profiles. Form Fig. 12(b), it can be seen that as the approximated degradation signal from the proposed approach reflects the changes in the operating profile, it is very close to the actual degradation signal. While the approximated degradation signal from the traditional approach is very far from the actual degradation signal, not reflecting the changes induced by the dynamic operating profiles. As a result, the RUL approximated

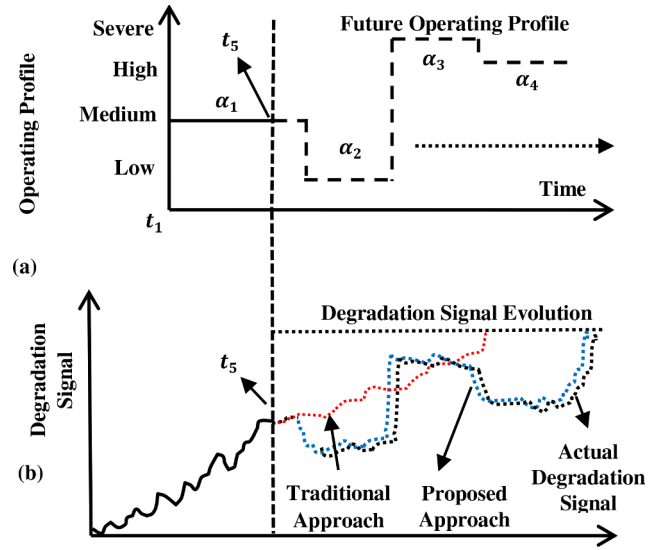


Fig. 12. Traditional approach/way vs proposed approach.

from the proposed approach, in contrast to the traditional approach, is more close to the actual RUL. This assessment is extensively distinguished experimentally as follows:

**Accuracy:** The prediction accuracy (PA) is the degree of the RULs correctly predicted amongst the total number of observations assessed. The results (see, Table 3) manifest that under all the scenarios our approach has shown improved performance, with PA resulting 89.60% in contrast to 79.96% from the traditional approach for scenario I. Whereas, for scenario II and III, the PA attained was more than 90% which is a significant improvement over 47.79 and 30.54 from the traditional approaches.

$$PA = \frac{1}{N} \sum_{i=1}^N \left[ 1 - \left( \frac{|RUL_{a_i} - RUL_{p_i}|}{RUL_{a_i}} \right) \right] \times 100. \quad (19)$$

where  $N$  is the total number of observations,  $RUL_{p_i}$  is the predicted RUL, and  $RUL_{a_i}$  is the actual RUL.

**Suitability:** For this, mean absolute error (MAE) is calculated. MAE measures how the system makes close RUL predictions to the actual

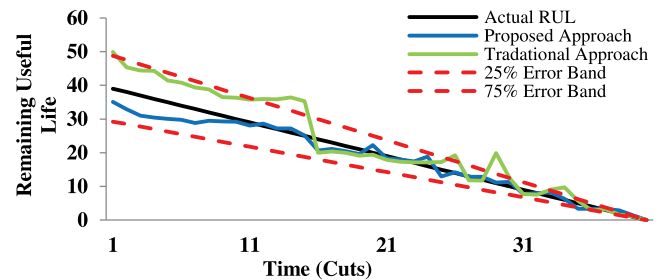


Fig. 13. Error bands on the predictions in the scenario I.

**Table 3**  
Results of comparative analysis.

| Industrial scenarios |                      | I                 |                      | II                |                      | III               |                      |
|----------------------|----------------------|-------------------|----------------------|-------------------|----------------------|-------------------|----------------------|
| Performance metric   |                      | Proposed approach | Traditional approach | Proposed approach | Traditional approach | Proposed approach | Traditional approach |
| Accuracy             | PA (%)               | 89.60             | 79.96                | 90.23             | 47.79                | 93.31             | 30.54                |
| Suitability          | MAE                  | 1.691             | 4.033                | 1.864             | 7.821                | 0.604             | 4.843                |
| Stability            | RAE (%)              | 16.91             | 40.33                | 20.18             | 84.61                | 10.07             | 80.71                |
|                      | RRSE (%)             | 20.60             | 45.84                | 24.98             | 83.16                | 13.92             | 111.94               |
| Quality              | R <sup>2</sup>       | 0.976             | 0.947                | 0.965             | 0.333                | 0.983             | 0.878                |
| Reliability          | RMSE                 | 1.300             | 2.008                | 1.365             | 2.797                | 0.777             | 2.201                |
| Applicability        | Computation time (s) | 0.440             | 0.425                | 0.312             | 0.225                | 0.455             | 0.446                |

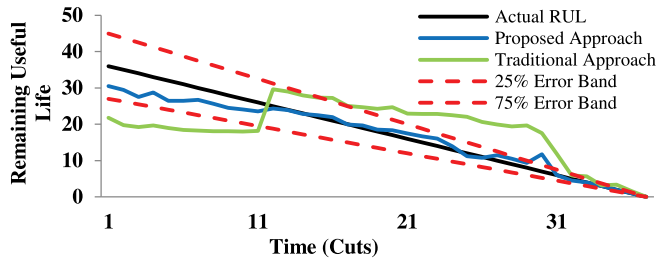


Fig. 14. Error bands on the predictions in scenario II.

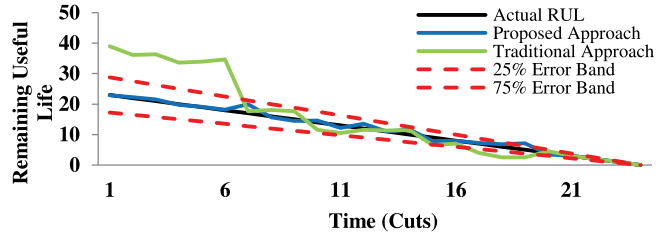


Fig. 15. Error bands on the predictions in scenario III.

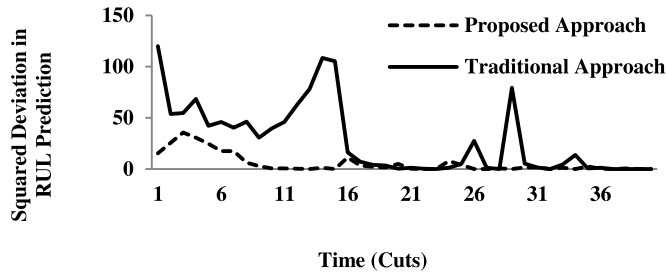


Fig. 16. Comparative results of squared deviation in the scenario I.

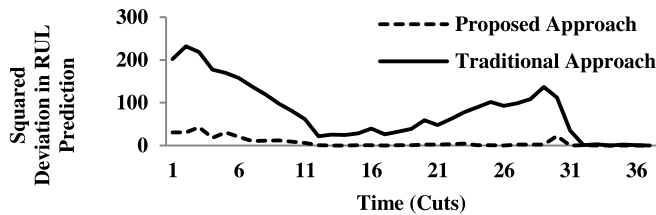


Fig. 17. Comparative results of squared deviation in scenario II.

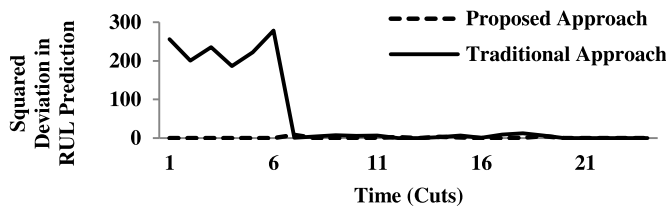


Fig. 18. Comparative results of squared deviation in scenario III.

RUL; lower MAE is better, a perfect model would score zero MAE. The MAE values acquired from the traditional approach is found to be about three times higher than the values attained from the proposed approach. Lower MAE values from the offered approach prove the suitability of the system under time-variant industrial scenarios by predicting the RULs closer to the actual RULs.

**Stability:** For this, relative absolute error (RAE) and root relative squared error (RRSE) are evaluated; these are the measures of the variance in the predictions. The lower RAE and RRSE values from the offered approach signify the stability of the system under diverse

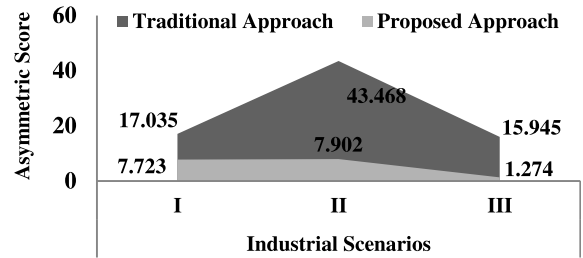


Fig. 19. Asymmetric score comparison under diverse scenarios.

industrial scenarios. For instance, in scenario II these errors are 76.15 and 69.96% lower than that attained from the traditional approach.

**Quality:** This is assessed through the goodness of fit. For which R-squared ( $R^2$ ) correlation coefficient is calculated. The high  $R^2$  correlation coefficient from the proposed approach displays that anticipated RULs are greatly associated with actual RULs in contrast to the traditional approach, showcasing better quality predictions from the offered approach.

**Reliability:** Root mean squared error (RMSE) is selected to indicate the reliability of the predictions; it illustrates the standard deviation of the differences between predicted and actual RULs; RMSE closer to zero indicates greater reliability. The higher reliability of RUL predictions under time-variant scenarios from the proposed approach is apparent, as RMSE values are two times lesser than the traditional approach.

**Robustness:** The robustness is visualized by plotting the error bands [35]. Herein, the bands for the RUL estimates are 25% deviations from the actual RULs in either direction. Lesser the deviations from the actual RULs, better is the performance. As a result, error bands make it easier to interpret the robustness of the approach. It can be observed from Figs. 13–15 that under different scenarios, from beginning to end of the operational life of tools, almost all values of the estimated RULs from the proposed approach lie within the specified error bands. Whereas, estimated RULs from the traditional approach in the beginning mostly lies outside the error bands and only lies within the bands at the end of operational life. It can be summarized that the proposed approach offers accurate prediction even in the early operation life. This is highly desirable for planning maintenance at the early stage of life. This performance suggests that the proposed approach makes reasonably robust estimates of the RUL under dynamic operating profiles at all times.

**Comprehensibility:** For this, the squares of the deviations around the RULs allied with these approaches are computed. The progressing trails of the squared deviation at each observation for both do under diverse scenarios are shown in Figs. 16–18. As the offered approach effusively exploits the future characteristics of dynamic operating profiles, the squared deviation of the RUL approximation is maintained at a reasonably low level in contrast to the traditional approach under all scenarios.

**Applicability:** In real-world applications, it is always desired to foresee failures early than late. Hence, to deliver a more valued and overall valuation of prediction performance, an asymmetric score metric (see, (20)) is computed. This score stresses substantial penalties on late prediction than an early prediction, and the penalty propagates exponentially with an increasing error. It is evident from Fig. 19 that the offered approach outperforms the traditional approach in all scenarios by prompting a lower score. This again confirms the findings from Figs. 13–15. This greater extent of improvement in performance under time-variant industrial scenarios is resultant of uniting the effects of the future characteristics of dynamic operating profiles in the proposed approach.

$$\text{Asymmetric Score} = \begin{cases} \sum_{i=1}^N e^{\left(\frac{RUL_{p_i} - RUL_{a_i}}{10}\right)} - 1 & \text{for } (RUL_{p_i} - RUL_{a_i}) < 0 \\ \sum_{i=1}^N e^{\left(\frac{RUL_{p_i} - RUL_{a_i}}{13}\right)} - 1 & \text{for } (RUL_{p_i} - RUL_{a_i}) \geq 0 \end{cases} \quad (20)$$

where  $N$  is the total number of observations,  $RUL_{p_i}$  is the predicted RUL, and  $RUL_{a_i}$  is the actual RUL.

In essence, all these implementation results lend significant credibility to the appropriateness of our approach over the traditional approach.

## 5. Conclusion

Current work progresses the existing body of knowledge by formulating a novel and generic TCM system for dynamic operating profiles. The purpose of this work was to equip machining industries with intelligence that allows responding to the time-variant operating profiles and adaptable under various real-world scenarios. The core contributions are highlighted as follows:

A realistic mathematical framework via a new, adaptive, and hybrid stochastic degradation model is offered, that inventively models degradation signal of tools functioning under dynamic profile, to prognosticate the RULs under diverse industrial scenarios.

To precisely approximate the degradation under dynamic profiles, the framework uniquely leverages strategic information viz. the evolution of the future profile, jerks owing to dynamic transitions, etc. Also, to take along realistic characteristics, new mappings, i.e., degradation rate function, and jerk function are formulated. As well, a profile sorting algorithm is offered.

For the first time, we innovatively modeled the physics of evolution of dynamic profiles to incite generalization in diverse real-world scenarios viz. batch production, job production, micro to medium-scale

## Appendix A

We describe the method for estimating the unknown parameter ( $\sigma$ ) of the white noise stochastic process from real-life data utilizing maximum likelihood estimation technique. Accordingly,  $\varepsilon(t)$  for different time instances is as follows:

$$\varepsilon(t) = D_s(t) - D_s(0) - \varphi(\omega(\xi(x))) - \sum_{j=1}^{Y(t)} G(\xi(Z_j)).$$

Next, we first concatenate  $\varepsilon(t)$  for different time instances, say,  $\{t_1, \dots, t_N\}$  in a vector  $\boldsymbol{\varepsilon} = \{\varepsilon(t_1), \varepsilon(t_2), \dots, \varepsilon(t_N)\}$ , and formulate the likelihood function of  $\boldsymbol{\varepsilon}$ , given as follows:

$$L(\boldsymbol{\varepsilon}|\sigma^2) = \frac{1}{(\sqrt{2\pi\sigma^2})^N} \exp\left(-\frac{\sum_{i=1}^N \varepsilon^2(t_i)}{2\sigma^2}\right).$$

Taking the logarithm of  $L(\boldsymbol{\varepsilon}|\sigma^2)$  gives the log-likelihood function of  $\boldsymbol{\varepsilon}$ , which can be written as:

$$\ln L(\boldsymbol{\varepsilon}|\sigma^2) = \frac{N}{2} \ln \frac{1}{2\pi\sigma^2} - \frac{\sum_{i=1}^N \varepsilon^2(t_i)}{2\sigma^2}.$$

The maximum likelihood estimate (MLE) of  $\sigma^2$  is given by:

$$\hat{\sigma}_{MLE}^2 = \underset{\sigma^2}{\operatorname{argmax}} \ln L(\boldsymbol{\varepsilon}|\sigma^2).$$

Henceforth,  $\sigma^2$  which maximizes log-likelihood function  $\ln L(\boldsymbol{\varepsilon}|\sigma^2)$  is obtained by equating the first-order derivative of  $\ln L(\boldsymbol{\varepsilon}|\sigma^2)$  to zero. Taking the derivative of  $\ln L(\boldsymbol{\varepsilon}|\sigma^2)$  results in:

$$\frac{\partial}{\partial \sigma^2} \ln L(\boldsymbol{\varepsilon}|\sigma^2) = -\frac{N}{2\sigma^2} + \frac{\sum_{i=1}^N \varepsilon^2(t_i)}{2(\sigma^2)^2}.$$

Equating  $\frac{\partial}{\partial \sigma^2} \ln L(\boldsymbol{\varepsilon}|\sigma^2)$  to zero and solving for the parameter of white noise stochastic process ( $\sigma$ ) yields:

$$\hat{\sigma}_{MLE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \varepsilon^2(t_i)}.$$

production environments.

A real-life case study is exhibited to validate the system's qualitative and quantitative performance. Also, the performance is evaluated in contrast to the traditional approach. The results confirmed that the offered approach delivers a robust problem-solving structure.

In essence, this work will help enrich the existing systems to compute the RULs while exploiting prior information, along with the future characteristics of profiles that the tool is likely to experience.

In the view of experimental restrictions viz. time, life test length, etc. only two operating profiles with different combinations are considered in experimental verification. Although, the critical characteristic of the proposed model is that it is capable of considering any number of operating profiles and is not restricted to just two specific profiles. However, for more in-depth verification, more operating profiles and their combinations can be considered.

## CRediT authorship contribution statement

**Amit Kumar Jain:** Conceptualization, Methodology, Software, Investigation, Validation, Writing - original draft. **Bhupesh Kumar Lad:** Conceptualization, Supervision, Funding acquisition, Writing - review & editing.

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